Randomized ZKP Cryptography with Zero-Knowledge

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Abstract

In the modern world, the authentication process to access application services is essential, but it sometimes results in the collection of unnecessary personal information from users. To prevent this, Zero Knowledge Proof (ZKP) cryptography is used, enabling a secure and efficient authentication process that does not violate privacy laws. To make it even more secure, randomness is added to store the entered personal information in blocks, and the block values are sent randomly during transmission, which is the main advantage of the Random Zero Knowledge Proof cryptographic technique.

Index Terms: Zero Knowledge Proofs. Block Chain, Decentralization, Random, User Authentication

I. INTRODUCTION

In the modern world, the process of authenticating with personal information is essential to access any site or print a document[3].

In the past, users tried to authenticate using ID and password, but nowadays, users can authenticate in various ways, such as simple authentication, financial certificate, i-pin, etc.

Although these various methods allow for quick authentication, the process of processing this data requires the input of unnecessary personal information, which reduces the safety of providing application services, and research has been focused on ways to improve it. One of the methods that can be applied to this problem is the ZKP cryptographic technique[2][7].

To briefly explain the use of ZKP, if a user's age needs to be verified, it can be done through an ID card. However, the problem with IDs is that in addition to age, they also contain other personal information such as address and gender. To solve this problem, the ZKP cryptographic technique can be used to protect the user's privacy by hiding the detailed personal information and only providing information about the age.

This paper will describe ZKP as one of the ways to protect the user's privacy during the authentication process, explain what algorithms are used to implement the concept, the

benefits and applications of this technique, and conclude with the fact that it can solve the problem of confidentiality of transactions, which is a problem in the current era[1].

II. ZKP

A. Overview of ZKP

Zero-Knowledge Proof (ZKP) is an important concept in cryptography, which refers to a cryptographic technique that allows a user to prove that they possess information without having to provide it directly. This cryptographic technique is used very effectively to authenticate personal information and is an important aspect of blockchain technology and cryptography[1].

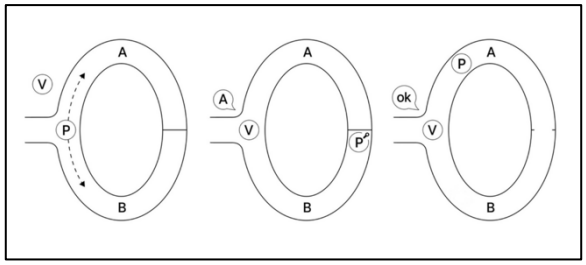

Fig. 1. ZKP authentication technique
(https://www.coindeskkorea.com/news/articleView.html?idxno=79339)

There are three main elements to the ZKP cryptographic technique. The first is Completeness, where the verifier believes the prover's performance to be true if the given condition is true. The second is Soundness, in which the prover cannot deceive the verifier if the given condition is False. Finally, the third is Zero-Knowledge, where the prover cannot know anything other than True or False[4][6][9].

Depending on the computational power of the prover, the methods are divided into Proofs and Arguments. A proof is called a proof if it satisfies soundness for a prover with infinite processing power. Proofs are called Arguments if they satisfy soundness for a prover with limited processing power.

Currently, there are three main types of ZKP cryptographic techniques: classification by initialization, classification by efficiency measure, and classification by post-quantum resistance[5][10].

B. Algorithm

By utilizing the Random Algorithm, which can add randomness to the process of the existing ZKP cryptographic technique, it will be possible to solve the problem of personal information leakage by randomizing the user's personal information so that only the information required for authentication can be provided[11].

Among the various Random Algorithms, the Shuffle Algorithm directly manipulates the original array, and at each step, one of the remaining elements is randomly selected

and exchanged with the current position, and the process is repeated for all elements to create a randomly shuffled array. The Shuffle Algorithm is how the first element of the resulting array is selected. This algorithm has a memory complexity of O(1) and a time complexity of O(n) because it does not require an additional array[12].

III. Suggestion method

To make the ZKP authentication technique more secure and efficient, we will describe an enhanced ZKP process in Chapter 3 that utilizes zero-knowledge to protect unnecessary information during user authentication in the real world.

A. ZKP with randomness

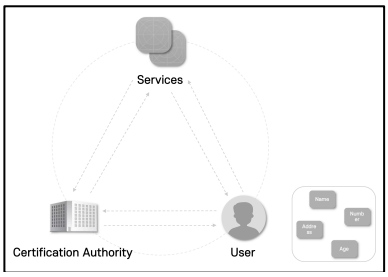

Fig. 2. Random ZKP authentication technique

[Assumption 1] Specify the user's private information as a single block.

[Assumption 2] Specify the block value of the user's private information to be Random for the proof.

B. ZKP's revamped authentication process

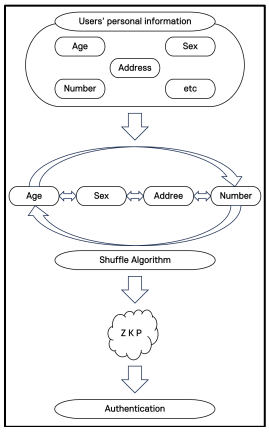

Fig. 3. Random ZKP encryption process

① The user accesses the application service and the service requests personal information from the user.

② The user enters personal information

The personal information entered through [Assumption 1] is stored as a block.

③ The stored block values are sent to the certification authority for verification.

Through [Assumption 2], randomly designated blocks are sent to the certification authority, and it is judged that the block value sent is unknown, which can increase safety and efficiency.

④ The certification authority verifies the block value and passes this information to the application service.

⑤ The application service compares the randomly received blocks to authenticate the user.

⑥ If they match, authentication is complete and the user can use the application service.

C. Improved ZKP

As described in Section 3.2, the main feature of the improved ZKP is that the user's personal information is stored in blocks, and the blocks are randomly selected and transmitted.

If a hacker attempts an attack to steal your personal information, the randomness of the improved ZKP makes it difficult for them to find out your entire personal information. In other words, randomness can increase security by making it difficult to identify patterns in the data. Here, even if a hacker manages to get a partial block value, the blockchain's features prevent them from knowing all of your personal information.

For example, even if the user entered the information in order, such as name, age, and gender, it would be difficult to determine the overall flow or structure of the data due to randomness.

IV. Conclusion

With the advancement of technology, users are receiving services provided by companies and organizations online. To avail of these services, users need to authenticate their personal information online and manage their verified IDs and passwords.

However, as this industry grows, the risk of users' personal information leakage exists and the potential threats have increased.

For instance, in 2023, there was a case where approximately 290,000 people's names, birth dates, and mobile phone numbers were leaked due to a personal information leakage controversy at LG U+.

Therefore, it is necessary to introduce a method that can protect personal information and conveniently verify as needed.

By using randomness ZKP (Zero-Knowledge Proof), if users can prove to the server that they know the password without transmitting it, there is no risk of personal information being exposed.

In the effort to prevent such personal information leakage, ZKP (Zero-Knowledge Proof) encryption technique is considered one of the important cryptographic

The randomness ZKP encryption technique we have conceived designates each piece of user's personal information as individual blocks within the existing ZKP encryption method,

and assigns randomness during the transmission process for verification. We believe this is one of the methods that can enhance efficiency and safety. It remains an ongoing task that requires continuous improvement and future research to concretely commercialize this.

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